

Public cohorts (GEO)

GSE224093

Severe IUA 7 vs Control 7
bulk RNA-seq (FPKM)

GSE215968

AS 40,336 vs WOI 66,064 cells
scRNA-seq, 17 cell types

GSE311899

9 AS patients, Pre/Post
CD133+ therapy, Epi/Stroma

GSE160633

Thin vs adjacent normal
8 IUA patients (direction)

Analysis

Bulk scoring

rank-based pathway scores
Ferroptosis / HMGB1 / Notch / Wnt
DEG direction + Spearman

Single-cell modules

per-cell z-score, 4 modules
activity / defense / iron / inflammation
MWU+BH per cell type
+ sample-level robustness

Findings

Defense collapse, not execution

GPX4/DHODH/KEAP1 axis down; score lower

Epithelium = dual hit

activity +0.145 / defense -0.244

Macrophages = iron sink

iron load +0.163, highest of all types

Myeloid-anchored coupling

DC 0.31 > Mac 0.30 >> Epi 0.07

HMGB1 redistribution

parenchyma down / myeloid up

Therapeutic implication: restore epithelial ferroptosis defense — susceptible state is potentially reversible;
myeloid iron niche shared across fibrotic pregnancy disease (IUA / PE)